Post-perturbation held-out DCLM loss

30M (0.03B) — adamw

30M (0.03B) — muon

60M (0.06B) — adamw

60M (0.06B) — muon

100M (0.1B) — adamw

100M (0.1B) — muon

300M (0.3B) — adamw

300M (0.3B) — muon

600M (0.6B) — adamw

600M (0.6B) — muon

DCLM_heldout loss

chinchilla (token budget)

γ=0.003 γ=0.005 γ=0.007 γ=0.01 γ=0.02 γ=0.03 γ=0.04 γ=0.05 γ=0.07 γ=0.1 γ=0.13 unperturbed